

Graph on the Coordinate Plane

Flagship

Lesson 9-1-flagship

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

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LOST COVE MISSION

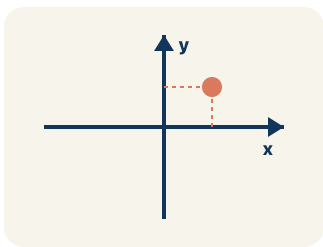
**Captain Vega's Coordinates**

You are the navigator on Captain Vega's expedition. Instead of 'X marks the spot,' she hid the treasure route as ordered pairs scrawled across a four-quadrant chart. Plot every point in the right order and the lost cove treasure is yours.

## Key Vocabulary

Level 1 support

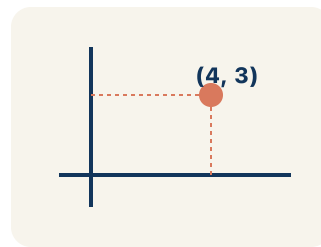
Picture first, then the word, then a plain-language meaning. Say each word out loud.



*A grid with a horizontal line (x-axis) crossing a vertical line (y-axis), making four sections*

### Coordinate plane

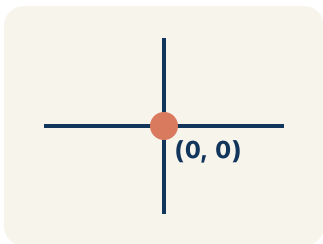
A grid with a line going across and a line going up that cross.



*(3, 5) means move right 3 from the origin, then up 5*

### Ordered pair

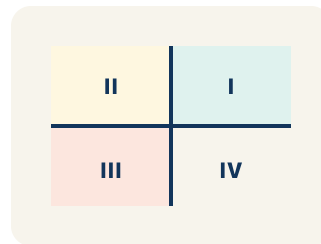
Two numbers (x, y) that tell where a point is on a grid.



*The starting point at the center where both axes cross — (0, 0)*

### Origin

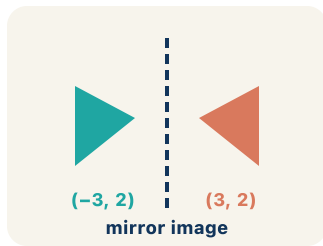
The point (0, 0) where the two grid lines cross.



*I (+,+) top-right, II (-,+) top-left, III (-,-) bottom-left, IV (+,-) bottom-right*

### Quadrant

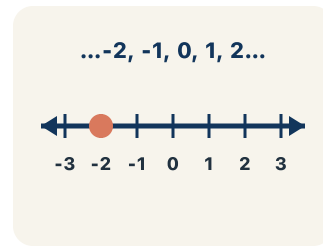
One of the four parts of a coordinate grid.



*$(3, 2)$  reflected over the  $x$ -axis becomes  $(3, -2)$  — same  $x$ , opposite  $y$*

## Reflection

A flipped, mirror image across a line.



## Integer

Whole numbers and their opposites, like -2, -1, 0, 1, 2.

## Key Ideas & Notes



### WHAT WE'RE LEARNING TODAY

**I can plot and identify points on the coordinate plane using ordered pairs.**

#### 1 Notice

Captain Vega left a treasure map, but instead of an 'X marks the spot,' she hid the directions as ordered pairs!

#### 2 Key idea

I can plot and identify points on the coordinate plane using ordered pairs.

#### 3 Words I'll use

Coordinate plane

Ordered pair

Origin

Quadrant

Reflection

Integer



#### Think About It

- What shape will the three points make when connected?
- Which number tells you how far right/left? Which tells you up/down?
- Why is the ORDER in an ordered pair important?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



#### I do — watch

Follow each step as your teacher solves it.

**Problem:** Which ordered pair names a point 3 units right and 7 units up from the origin?

- A. (3, 7)
- B. (7, 3)
- C. (3, 3)
- D. (0, 7)

**Step 1**

The first coordinate is horizontal (right 3), the second is vertical (up 7).

**Step 2**

So (3, 7).



**Answer:** A. (3, 7)

 **We do — together**

Solve this one with your class using the same steps.

**Problem:** What are the coordinates of the origin?

A. (0, 0)

B. (1, 1)

C. (0, 1)

D. (1, 0)

**Step 1**

\_\_\_\_\_

**Step 2**

\_\_\_\_\_

**Answer:**

\_\_\_\_\_

## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**On a treasure map laid over a coordinate plane, why do we always start at the origin (0, 0) and move sideways before moving up or down to reach the treasure?**

#### Level 1 support

**Start here:** We start at the origin and move along the x-axis first because \_\_\_\_.

 **Need a hint?**

**Sentence starters (optional)**

**The first number in (x, y) tells me to move \_\_\_\_.**

*El primer numero en (x, y) me dice que me mueva \_\_\_\_.*

**The second number tells me to move \_\_\_\_.**

*El segundo numero me dice que me mueva \_\_\_\_.*

**Word bank:** coordinate plane origin ordered pair x-axis y-axis

**Listen for:** Student states the x-coordinate is the sideways move and the y-coordinate is the up/down move, both starting from the origin.

#### Level 2

**Push students to explain why the treasure spots (3, 5) and (5, 3) are different locations even though they use the same numbers.**

- (3, 5) and (5, 3) are different spots because \_\_\_\_.
- The order in an ordered pair matters because \_\_\_\_.

**EXPLORE**

**Your map says the treasure is at (4, 2). Walk your partner through every move you make from the origin, using the words x-axis and y-axis.**

**Level 1 support**

**Start here:** Starting at the origin, I first move \_\_\_\_, then I move \_\_\_\_.



**Need a hint?**

**Sentence starters (optional)**

**I move \_\_\_\_ units along the x-axis.**

*Me muevo \_\_\_\_ unidades por el eje x.*

**Then I move \_\_\_\_ units along the y-axis.**

*Luego me muevo \_\_\_\_ unidades por el eje y.*

**Word bank:** **origin** **ordered pair** **x-axis** **y-axis** **coordinate plane**

**Listen for:** Student moves 4 right along the x-axis and 2 up along the y-axis, naming each axis and the origin as the start.

**Level 2**

**Ask students to predict which quadrant the treasure at (4, 2) lands in and justify using the signs of the coordinates.**

- The treasure at (4, 2) lands in quadrant \_\_\_\_ because \_\_\_\_.
- I can tell the quadrant from the coordinates by \_\_\_\_.

CONNECT

**How is reading a coordinate pair on the treasure map like reading directions in a video game or on a real map?**

Level 1 support

**Start here:** An ordered pair works like map directions because \_\_\_\_.



**Need a hint?**

**Sentence starters (optional)**

**The x-coordinate is like going \_\_\_\_.**

*La coordenada x es como ir \_\_\_\_.*

**The y-coordinate is like going \_\_\_\_.**

*La coordenada y es como ir \_\_\_\_.*

**Word bank:** **ordered pair** **coordinate plane** **x-axis** **y-axis** **origin**

**Listen for:** Student connects the *x*-value to horizontal travel and the *y*-value to vertical travel, like map directions from a start point.

Level 2

**Push students to generalize a rule for plotting ANY treasure location, including ones with a zero coordinate.**

- A rule that always works for plotting  $(x, y)$  is \_\_\_\_.
- If one coordinate is 0, the treasure lands on \_\_\_\_ because \_\_\_\_.

REFLECT

A teammate marked the treasure at (2, 6) by moving up 2 and over 6. What mistake did they make, and how would you coach them?

Level 1 support

**Start here:** The mistake was \_\_\_\_, and to fix it they should \_\_\_\_.



Need a hint?

**Sentence starters (optional)**

**They mixed up the \_\_\_\_ and the \_\_\_\_.**

*Confundieron la \_\_\_\_ y la \_\_\_\_.*

**The correct way is to move \_\_\_\_ first.**

*La forma correcta es moverse \_\_\_\_ primero.*

**Word bank:** ordered pair x-axis y-axis origin coordinate plane

**Listen for:** Student identifies that the  $x$ - and  $y$ -coordinates were swapped and explains the correct order ( $x$  first, then  $y$ ).

Level 2

**Ask students to critique why this swap error is so common on maps and suggest a strategy to avoid it.**

- This error happens a lot because \_\_\_\_.
- A strategy to remember the order is \_\_\_\_.

## Write About the Math

### The Writing Revolution

I can explain my work using the words coordinate plane, ordered pair, origin, and quadrant.

#### 1. Kernel Sentence subject + verb

**Model:** Coordinate plane is a grid with a line going across and a line going up that cross.

*Plano cartesiano es una cuadrícula con una línea horizontal y una vertical que se cruzan.*

**Write a kernel sentence about coordinate plane. Use a subject and a verb.**

*Escribe una oración base sobre plano cartesiano. Usa un sujeto y un verbo.*

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#### 2. Sentence Expansion because · but · so

**Kernel:** Coordinate plane matters in math

*Plano cartesiano importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Coordinate plane matters in math because \_\_\_\_.**

*Plano cartesiano importa en matemáticas porque \_\_\_\_.*

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**but**  
*pero*

**Coordinate plane matters in math, but \_\_\_\_.**

*Plano cartesiano importa en matemáticas, pero \_\_\_\_.*

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**so**  
*entonces*

**Coordinate plane matters in math, so \_\_\_\_.**

*Plano cartesiano importa en matemáticas, entonces \_\_\_\_.*

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### 3. Sentence Types 4 ways to write a math idea

**Statement**  
*Afirmación*

Tell one true fact about coordinate plane.  
*Di un hecho verdadero sobre coordinate plane.*

**Coordinate plane** \_\_\_\_.

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**Question**  
*Pregunta*

Ask a question about coordinate plane.  
*Haz una pregunta sobre coordinate plane.*

**How does** \_\_\_\_ ?

*¿Cómo* \_\_\_\_ ?

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**Exclamation**  
*Exclamación*

Show excitement about coordinate plane.  
*Muestra entusiasmo sobre coordinate plane.*

**Wow,** \_\_\_\_ !

*¡Guau,* \_\_\_\_ !

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**Command**  
*Mandato*

Tell a partner what to do with coordinate plane.  
*Dile a un compañero qué hacer con coordinate plane.*

**First,** \_\_\_\_ .

*Primero,* \_\_\_\_ .

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### 4. Explain Your Reasoning use a sentence starter

**The first number in (x, y) tells me to move** \_\_\_\_.

*El primer numero en (x, y) me dice que me mueva* \_\_\_\_.

**The second number tells me to move** \_\_\_\_.

*El segundo numero me dice que me mueva* \_\_\_\_.

**I move** \_\_\_\_ **units along the x-axis.**

*Me muevo* \_\_\_\_ *unidades por el eje x.*

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## Try It

Solve on your own. Check the answer key when you are done.

**1. A point is at (6, 2). What does the 6 tell you?**

- A. Move 6 units to the right
- B. Move 6 units up
- C. Move 6 units to the left
- D. The point is in Quadrant 6

Show your work:

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### Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**A student says that (3, 5) and (5, 3) are the same point because they use the same numbers. Is the student correct? Explain why or why not, and describe where each point is located on the coordinate plane.**

*Sentence starter: The student is \_\_\_\_ because (3, 5) means \_\_\_\_ while (5, 3) means \_\_\_\_\_. The ORDER matters because \_\_\_\_\_.*

Show your work:

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## Reflect — Exit Ticket

**Point P is located 6 units right and 3 units up from the origin. What ordered pair names point P?**

- A. (6, 3)
- B. (3, 6)
- C. (6, 0)
- D. (0, 3)

Your answer:

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## Answer Key & Teacher Guide

1. **We Try (notes):** A.  $(0, 0)$  — *The origin is where the x-axis and y-axis cross, at  $(0, 0)$ .*
2. **Try It 1:** A. Move 6 units to the right — *The first number (x-coordinate) tells you horizontal movement. 6 means go right 6 units from the origin.*
3. **Exit Ticket:** A.  $(6, 3)$  — *Right 6 = x-coordinate of 6, Up 3 = y-coordinate of 3. The ordered pair is  $(6, 3)$ .*

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Coordinate plane is a grid with a line going across and a line going up that cross.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).